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Calibrating autonomous recording units to estimate sound pressure levels using frequency-band univariate models and microphone sensitivity

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HIGHLIGHTS

- Field-paired models convert ARU octave-band levels to meter-referenced SPL.
- Ten-fold cross-validation yielded 3.77 dB MAE and 4.82 dB RMSE.
- Device type and frequency band strongly affected calibration accuracy.
- Smartphones showed larger errors than purpose-built autonomous recorders.
- Coefficients apply within the sampled devices, conditions, and workflow.

ARTICLE INFO

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ABSTRACT

Anthropogenic sound is widespread across human-dominated and semi-natural landscapes, and many applications require spatially dense exposure estimates that are difficult to obtain with sound level meters alone. Autonomous recording units (ARUs) enable dense acoustic sampling, but audio-derived levels are device- and workflow-dependent and therefore require empirical calibration before they can be compared with meter-referenced sound pressure levels (SPL). Here we derive field-paired octave-band calibration equations that map 60 s ARU-derived band levels to concurrent sound level meter measurements across seven field sound contexts. Using univariate, band-specific models with device-type offsets, unit microphone sensitivity, and meteorological covariates, we show that recorder-derived levels can be translated into meter-referenced octave-band SPL with quantified uncertainty within the sampled domain. Across 10-fold cross-validation, overall bias was −0.07 dB, mean absolute error was 3.77 dB, and root mean squared error was 4.82 dB; 72% of estimates were within ±5 dB of the meter. Performance varied by device and band, with Song Meter units clustering between 4.37 and 4.66 dB RMSE, smartphones showing larger errors (6.08 dB RMSE), and larger errors at mid-frequency bands and high-SPL observations. We provide coefficient tables, diagnostics, and worked examples for applying the calibration to new 60 s recordings from tested device types processed with the same extraction workflow. The method offers a reproducible route from ARU recordings to meter-referenced octave-band SPL estimates under defined conditions, while retaining explicit limits on extrapolation beyond the sampled devices, weather, sound contexts, and signal-processing pipeline.

1. Introduction

Anthropogenic sound is widespread across human-dominated and semi-natural landscapes, and it can alter animal behaviour, communication, and habitat use; it can also be treated as an exposure of interest in environmental assessment [1–3]. Exposure assessment requires measurements that are comparable across sites and time, and this comparability is often anchored to standardised sound pressure level metrics measured by sound level meters. In practice, instrument cost and operational constraints limit the spatial density and duration of conventional sound level meter deployments. Calibrated ARU networks can extend coverage by providing meter-referenced estimates at many more locations, provided the calibration is explicit, frequency-aware, and portable. Calibrated ARUs also provide the additional benefit

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of simultaneously collecting information on wildlife occurrence and acoustic behaviour, including calling rate, call amplitude, and, when callers can be localised, source-level vocal amplitude, which can improve assessment of how anthropogenic sound affects animals. This paper develops field-based calibration equations that translate octave-band levels extracted from autonomous recording units (ARUs) into octave-band sound pressure levels (SPL; dB re 20 μPa) measured by a reference sound level meter under matched field conditions. ARUs enable dense spatial sampling and long deployments in ecological acoustics, but recorder-derived levels are device- and workflow-dependent and therefore require explicit empirical calibration before they can be compared with meter-referenced exposure summaries. By modelling octave bands one at a time and retaining device type and microphone sensitivity as covariates, we provide coefficients that can be applied directly to recordings processed with the same workflow to obtain meter-referenced octave-band SPL estimates with quantified error. This supports consistent, reproducible estimation of sound exposure from remotely deployed recorders when the application remains within the device types, processing choices, and environmental domain represented by the calibration data. Prior work has demonstrated that recorders differ in sensitivity, self-noise, and spectral response, and that recorder-to-meter calibration can be expressed as frequency-dependent relationships [4]. These device and frequency dependencies imply that pooled conversions can misrepresent exposure if recorder identity and spectral structure are ignored. We therefore use paired field sampling spanning multiple sound contexts and fit univariate, band-specific calibration relationships with explicit device-type terms and a unit microphone-sensitivity covariate. The result is a set of coefficient tables and an application recipe that translate recorder-derived octave-band summaries into meter-referenced octave-band SPL for new recordings processed with the same extraction pipeline.

2. Methods

This section describes the paired field design, the extraction of minute-average octave-band summaries from sound level meter logs and co-located audio recordings, and the univariate calibration model used to estimate meter-referenced SPL. Calibration requires measurements that are matched in time and context, and it also requires that octave-band definitions and covariates are defined consistently across devices. The protocol yields per-band linear equations that can be applied to new 60 s recordings processed with the same spectral summarisation choices. This supports routine estimation of octave-band SPL from ARU deployments using device-specific calibration.

2.1. Field contexts and paired measurements

We sampled seven real-world contexts: airport, backyard, construction, industrial, light rail transit, park, and road. These contexts were chosen to span typical anthropogenic and mixed soundscapes because empirical calibration for field applications benefits from exposure to diverse spectral and temporal structures rather than reliance on a single source. Within each context we deployed three independent replicate “strips” (Fig. 1) and recorded each strip as a sequence of consecutive 60 s observations. This design provides replicated, paired measurements while keeping the unit of analysis aligned with common monitoring situations.

We define one observation as one 60 s window for which the reference meter produced minute-average octave-band SPL and each device produced a co-timed 60 s audio recording summarised to octave-band predictors. Minute averages reduce sensitivity to small start-time offsets across instruments and stabilise summaries relative to second-scale variability, which is typically not the target of landscape-scale mapping. The resulting dataset comprised 64 minute-level windows, yielding 380 device-minutes and 3800 paired device-band observations across 10 octave bands. All coefficients and performance results reported below are conditional on this minute-average observation definition.



Fig. 1. Example strip arrangement used for paired recordings. Devices were arranged in a fixed order and recorded in consecutive 60 s windows, while the reference SPLM logged octave-band SPL for the corresponding sequence of minutes.

2.2. Devices, units, and microphone sensitivity

Device types included Wildlife Acoustics Song Meter units (SM2, SM3, SM4, and SM Mini), Frontier Labs BAR-LT recorders, and Ulefone Armor 8 smartphones [5], and were treated as distinct device types in modelling and reporting. Device types differ in microphone characteristics and internal audio handling, which can produce differences in recorder-derived spectral summaries. For each device type we used three independent units, one per replicate strip, and we retained the unit-specific microphone sensitivity value recorded for each unit as a device-level covariate. Microphone sensitivity values were measured in the workshop prior to deployment using reference sounds and a Class 1 acoustic calibrator. This supports calibration to account for unit-to-unit sensitivity variation while keeping coefficients directly usable with recorded covariate values.

Smartphone recordings were collected through the ordinary device recording pathway, without specialised low-level or raw audio access and without disabling operating-system or device-level gain control, compression, or speech-oriented audio processing. This documents the acquisition pathway for the consumer-device component of the study and distinguishes it from a controlled electroacoustic characterisation of the phone hardware. Smartphones were therefore included as a direct-use comparison under the same field and processing workflows applied to the ARUs. Their results should be interpreted as the empirical performance for this practical acquisition pathway rather than as a claim about isolated smartphone microphone sensitivity.

The reference instrument was a Quest SoundPro® SE/DL series sound level meter (TSI Incorporated), configured for 1/1-octave real-time analysis and minute–average data logging to produce octave-band SPL (dB re 20 μPa) summaries used as calibration targets [6]. A reference SPLM provides an operational SPL scale against which recorder-derived summaries can be mapped. Our calibration targets are therefore meter-referenced octave-band SPL values for matched 60 s windows. The equations below should be interpreted as an empirically supported translation from recorder-derived predictors to meter-referenced SPL estimates.

2.3. Octave-band selection, meter logging, and audio summarisation

We modelled 10 one-octave bands with centre frequencies 16, 31.5, 63, 125, 250, 500, 1000, 2000, 4000, and 8000 Hz; each octave band spans a factor of two in frequency, with band edges at $f_c/\sqrt{2}$ and $f_c\sqrt{2}$ (for example, the 8 kHz band spans $\approx$ 5.66–11.31 kHz, and the 16 kHz band spans $\approx$ 11.31–22.63 kHz). Octave-band reporting preserves spectral information while avoiding an overly fine band partition that can be unnecessary for many field applications and can amplify instability

when data are limited at some frequencies. We excluded the 16 kHz octave band because many ARUs operate at a 32 kHz sampling rate (Nyquist frequency of 16 kHz), which means only the lower half of that octave band (11.31–16 kHz) would be observable and the upper half (16–22.63 kHz) would be absent, making the effective band definition inconsistent across devices and sampling-rate settings. The reported coefficients therefore apply to a consistent, device-comparable set of full octave bands up to 8 kHz.

SPLM octave-band SPL time series were exported from the SoundPro software as comma-separated value tables in long format, with repeated time stamps such that each logged minute contributes one row per octave band (that is, multiple rows per minute, one per band). This export structure requires reshaping to obtain a single minute-average SPL value for each band that can be paired with the corresponding 60 s audio window. We therefore grouped SPLM records by strip, minute index, and band, and used the mean SPL within each group as the meter-referenced response $y_{i,b}$ for observation i and band b. This yields one SPLM response per minute and band on the same temporal support as the device-derived predictors.

At the start of each strip, the operator produced a brief, clearly audible spoken cue ("start") within half a meter of the co-located instruments at the moment SPLM logging was initiated. Independent devices can differ slightly in start time, and aligning only by device clocks can misassign recorder windows to SPLM minutes. In post-processing, this cue was used as a common time origin, and SPLM data and device audio were then treated as consecutive 60 s windows from that origin and paired by minute index. This defines co-timed minute-average octave-band pairs within each strip in a way that is reproducible without requiring clock synchronisation across devices.

Device recordings were segmented into consecutive, non-overlapping 60 s windows and stored as WAV files, with segmentation performed in an audio editor [7]. A fixed 60 s window definition ensures that the samples contributing to each minute-average summary are unambiguous and match the SPLM's logging interval. All subsequent spectral summaries and model inputs were computed at the 60 s window scale. This keeps the calibration targets and predictors defined on the same temporal support.

Audio files were processed at their native sampling rate and bit depth as stored in the WAV header, and recordings could be mono or stereo depending on device configuration. Sampling rate constrains the observable frequency range, while channel structure can affect amplitude summaries if handled inconsistently. When recordings were stereo, we used a single-channel representation by selecting the channel containing the signal when one channel was effectively silent, otherwise by averaging channels. This yields a consistent one-channel waveform for spectral summarisation across devices.

For each 60 s window we estimated a Welch power spectral density (PSD) using Hann-windowed segments of length 4096 samples with 50% overlap [8,9]. Welch averaging provides a stable spectral summary for finite-duration recordings and reduces variance relative to a single periodogram, which is useful when deriving band-limited predictors from one-minute windows. Octave-band power was computed as the discrete integral of the one-sided PSD within each theoretical octave-band interval $(f_c/\sqrt{2}, f_c\sqrt{2})$, producing a band-power summary $P_{i,b}$ for observation i and band b. These band-power summaries provide a reproducible, frequency-specific predictor derived directly from the recorded waveform [4].

For modelling we used the log-transformed band-power predictor $x_{i,b} = 10\log_{10}(P_{i,b})$, retaining observations with finite recorder-derived band levels. Log scaling yields a decibel-like predictor that supports linear calibration while remaining explicit that the predictor is on a relative digital scale rather than an acoustic SPL scale. Accordingly, $x_{i,b}$ should be interpreted as a device-derived band level defined by the PSD integration and log transform above, and not as dB SPL or as a claimed dBFS measurement. The calibration model maps these recorder-derived predictors to meter-referenced SPL, and the resulting coefficients are conditional on this predictor definition [4,10].

Because the ARU predictor was computed by rectangular numerical integration of the Welch PSD, while the SPLM produced octave-band summaries through its internal analyser, we conducted a signal-processing sensitivity analysis on the calibration WAV files. This comparison quantifies how much the recorder-derived predictor changes when the rectangular integration step is replaced by a smooth roll-off octave-band approximation. For each calibration WAV file and octave band, we recomputed band power using cosine-tapered transition regions around the theoretical band edges, with the primary comparison using a transition width of 1/12 octave. The sensitivity calculation retained the 4096-sample Hann window and 50% overlap used in the main extraction workflow, but used zero-padding to a 65,536-point FFT to represent transition regions on a finer diagnostic frequency grid. This sensitivity analysis evaluates dependence on the stated extraction pipeline without treating the alternative calculation as a certified IEC 61260-1 filter implementation.

We also retained concurrent meteorological summaries for each observation, namely temperature, relative humidity, and wind speed, recorded using Kestrel 4000 Pocket Weather Tracker instruments [11]. Meteorological conditions can influence outdoor propagation and can covary with recorder performance over field deployments, which can otherwise appear as structured residual error in recorder-to-meter calibration [4]. For each strip, meteorology was recorded twice (start and end of the strip sequence) and the closest-in-time measurement was assigned to each minute observation. These covariates enter the calibration equation as additive adjustments, and the same adjustments can be applied when covariates are recorded and defined consistently with the calibration dataset.

2.4. Statistical model: univariate (per-band) device-specific calibration

We fit a univariate Gaussian calibration model in which each band has its own intercept, slope, and residual standard deviation, and device-type offsets and covariate effects are shared across bands. Band-specific intercepts and slopes capture frequency-dependent recorder-to-meter relationships, while shared terms stabilise estimation of device and covariate effects given a modest number of minutes. Let $y_{i,b}$ denote the reference SPLM octave-band SPL for observation i and band b, and let $x_{i,b}$ denote the corresponding recorder-derived predictor for the same minute window. Calibration then reduces to a per-band linear mapping with additive adjustments that can be applied directly.

$$y_{i,b} \sim \mathcal{N}\left(\mu_{i,b}, \sigma_b^2\right) \tag{1}$$

$$\mu_{i,b} = \alpha_b + \beta_b x_{i,b} + \gamma_{m_i} + \delta s_i + \theta_T T_i + \theta_H H_i + \theta_W W_i, \tag{2}$$

Device-type offsets were identified relative to BAR-LT, the first device level in the model input, by defining $\gamma_m = \gamma_m^{\mathrm{raw}} - \gamma_{\mathrm{BAR-LT}}^{\mathrm{raw}}$. This constraint is required because a model with both band-specific intercepts and unconstrained device-type offsets is otherwise not identifiable. The BAR-LT offset is therefore fixed at zero, and all other device-type offsets are interpreted as additive differences relative to BAR-LT. This defines the coefficient scale used for prediction and for the device-offset table below. Here m_i denotes device type, s_i is unit microphone sensitivity, and T_i, H_i, and W_i are temperature, relative humidity, and wind speed aligned to the observation definition above. This parameterisation matches the practical goal of plug-in prediction using recorded covariate values rather than effect-size comparison across predictors. We estimated posterior distributions for all coefficients using Markov chain Monte Carlo (MCMC) sampling implemented in `nimble` within R [12], and summarised coefficients using posterior medians and 95% credible intervals (CIs). Posterior summaries provide point estimates for application while retaining uncertainty that can be propagated in downstream analyses if required.

2.5. Cross-validation and performance metrics

We evaluated predictive performance using 10-fold cross-validation with folds assigned at the context-by-strip level, so that all device, minute, and band rows from a replicate strip were kept together. Strip-level grouping reduces information leakage that would occur if recordings from the same local deployment block contributed to both the training and test sets through different minutes, devices, or bands. For each fold we fitted the model to the training set and predicted SPLM SPL for the held-out strips, then aggregated errors across bands and device types. This yields out-of-sample error summaries that reflect how the calibration performs when applied to new deployment blocks drawn from the sampled contexts.

For each held-out prediction we defined the residual as $e = \mathrm{SPL}_{\mathrm{meter}} - \widehat{\mathrm{SPL}}$, so positive residuals indicate underestimation by the device-based estimate and negative residuals indicate overestimation. This sign convention is needed because bias, absolute error, and squared-error metrics summarise different aspects of the same residual distribution. We report bias as the mean of e, mean absolute error (MAE) as the mean of $|e|$, root mean squared error (RMSE) as the square root of the mean of e^2, median absolute error as the median of $|e|$, the proportion of estimates within ± 3 dB and ± 5 dB of the SPLM, and the proportion with absolute error at least 10 dB. We report these metrics overall, by device type, by octave band, and across the observed SPL range to identify where calibration is most reliable and where refinement is most beneficial.

2.6. Sensitivity analyses for meteorological covariates

To evaluate whether meteorological covariates were essential to predictive performance, we fit a no-weather sensitivity model that retained the same band-specific intercepts and slopes, device-type offsets, and microphone-sensitivity term, but removed temperature, relative humidity, and wind speed. This comparison addresses whether the submitted calibration depends strongly on meteorological adjustment within the narrow weather range sampled. The no-weather model provides a direct Bayesian sensitivity check against the primary weather model. This separates the operational value of the weather covariates from the broader question of whether they should be interpreted as physical estimates of weather-dependent microphone response.

We also fit a broad-context sensitivity model that replaced meteorological covariates with acoustic-context offsets for airport, backyard, construction, industrial, light rail transit, park, and road. This comparison evaluates whether the weather terms may partly absorb differences among acoustic contexts or deployment conditions. The context model was treated as a diagnostic alternative, not as the primary operational model, because new applications may not use the same context categories. Together, the no-weather and context sensitivity models provide a bounded interpretation of the primary weather model without changing the main calibration workflow.

3. Results

We first summarise dataset structure, then report out-of-sample performance under cross-validation, and finally provide the calibration coefficients required for application. Performance metrics quantify expected error, while coefficient tables provide the operational conversion required for reuse. Results are presented overall, by device type, and by octave band, followed by coefficient tables and a worked recipe. This ordering supports both interpretation and immediate application in new datasets.

3.1. Dataset summary

The paired dataset contained 64 minute-level windows across seven contexts, producing 380 device-minute recordings and 3800 paired device-band observations across 10 octave bands. This structure means each minute provides a shared SPLM reference against which multiple devices are evaluated under the same soundscape. Most minutes included the full set of device types, with a small number of minutes missing one device recording because a recorder occasionally began recording after SPLM logging had already started, yielding fewer paired device-band observations for that device in those minutes. Cross-validation grouped by context-by-strip preserves this structure in both training and evaluation by keeping all device, minute, and band rows from a replicate strip in the same fold.

Summary statistics for meteorological covariates (temperature, relative humidity, wind speed) and microphone sensitivity across the calibration dataset are provided in Appendix Tables C.1 and C.2.

3.2. Overall predictive performance

Across 10-fold cross-validation, overall bias was -0.07 dB, MAE was 3.77 dB, RMSE was 4.82 dB, and median absolute error was 3.12 dB. These summaries describe expected performance when applying the calibration to new minute-level recordings drawn from the same range of contexts. Approximately 49% of estimates fell within ± 3 dB and 72% within ± 5 dB of the SPLM, while 4.0% of estimates had absolute error at least 10 dB. These values provide realistic expectations for screening, mapping, and comparative exposure analyses based on calibrated octave-band summaries.

3.3. Residual structure across the SPL range

Cross-validated residuals were not constant across the full observed SPLM range (Fig. 2). This diagnostic evaluates whether the linear calibration captures recorder-to-meter relationships equally well at low, moderate, and high reference levels. Across all device types, observations above 70 dB were sparse and systematically underestimated, with positive residuals because $e = \mathrm{SPL}_{\mathrm{meter}} - \widehat{\mathrm{SPL}}$. The calibration is therefore best supported over the main sampled SPL range, while high-level events should be treated as a distinct domain requiring additional targeted calibration data.

Only 143 of 3800 held-out device-band predictions had SPLM values above 70 dB, and these high-SPL observations occurred in the 16, 31.5, 63, and 125 Hz bands. This concentration means that high-SPL behaviour was estimated primarily from low-frequency observations in airport, industrial, light rail transit, and road contexts. Prediction error increased in this region: observations from 70–75 dB had an RMSE of 7.98 dB, observations from 75–80 dB had an RMSE of 7.85 dB, and observations above 80 dB had an RMSE of 15.44 dB. These results show that the overall RMSE summarises the complete sampled dataset but does not describe a constant error level across the SPL range.

Recorder-derived octave-band predictors were also plotted against the reference SPLM SPL to evaluate the observed predictor-response relationship directly (Fig. 3). This diagnostic complements the residual plot by showing the field relationship between the audio-derived predictor and the meter-referenced response before the calibration error is summarised. The plot documents the empirical support for the fitted linear relationships across device types and the observed SPL range. Together with the residual diagnostics, it identifies the range where the calibration is best supported and where additional high-SPL calibration data would be most useful.

3.4. Performance by device type

Predictive accuracy differed by device type (Table 1), with smartphones showing the highest RMSE and a positive bias relative to the SPLM. Device-stratified summaries are needed because calibration is applied using the recorded device-type label. Dedicated ARUs had a combined RMSE of 4.52 dB, an MAE of 3.58 dB, and 2.5% of estimates with an absolute error of at least 10 dB, whereas smartphones had an RMSE of 6.08 dB, an MAE of 4.70 dB, and 11.4% of estimates with an absolute error of at least 10 dB. Device identity is therefore required for application, and smartphone predictions should be reported separately from purpose-built recorder predictions.

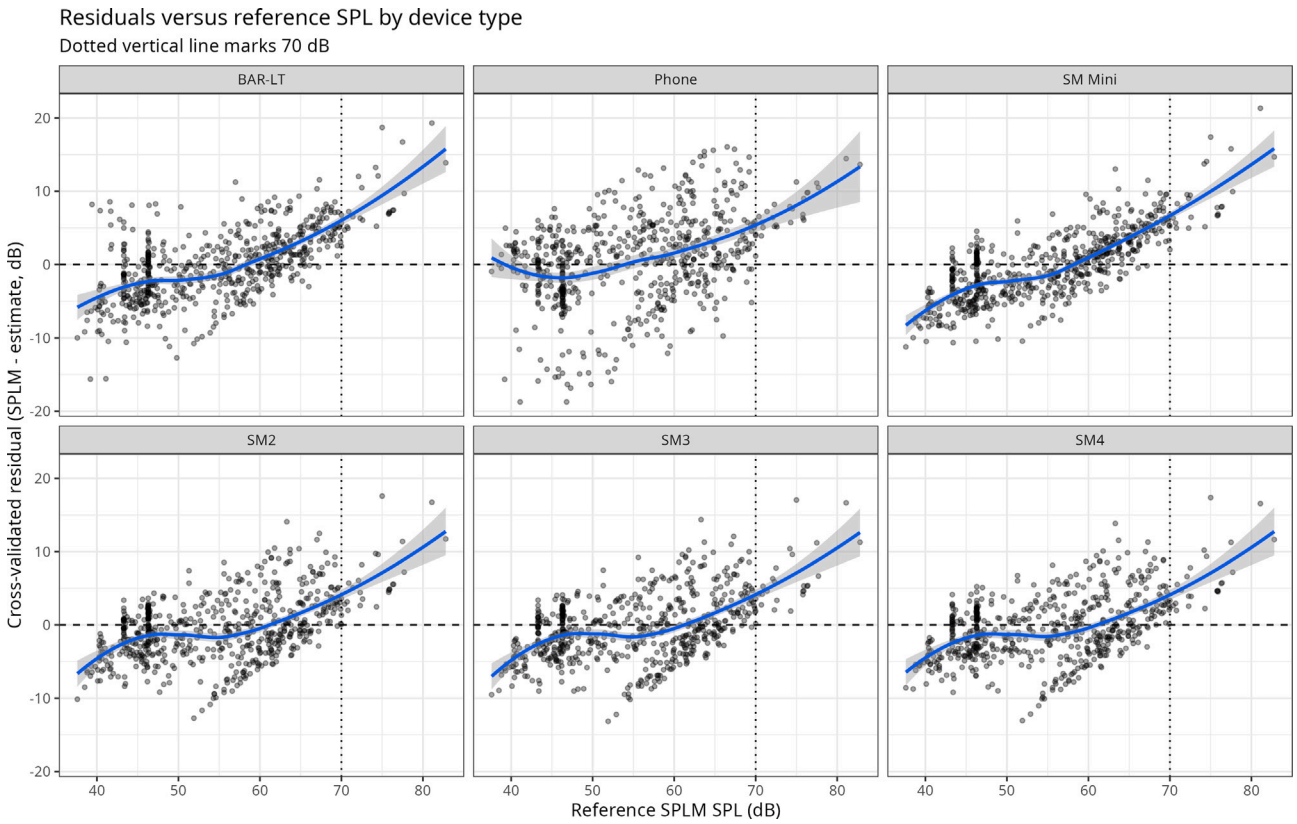


Fig. 2. Cross-validated residuals as a function of the reference SPLM SPL by device type. Residuals are defined as SPLM SPL minus the calibrated estimate, so positive values indicate underestimation by the device-based estimate. The vertical dotted line marks 70 dB.

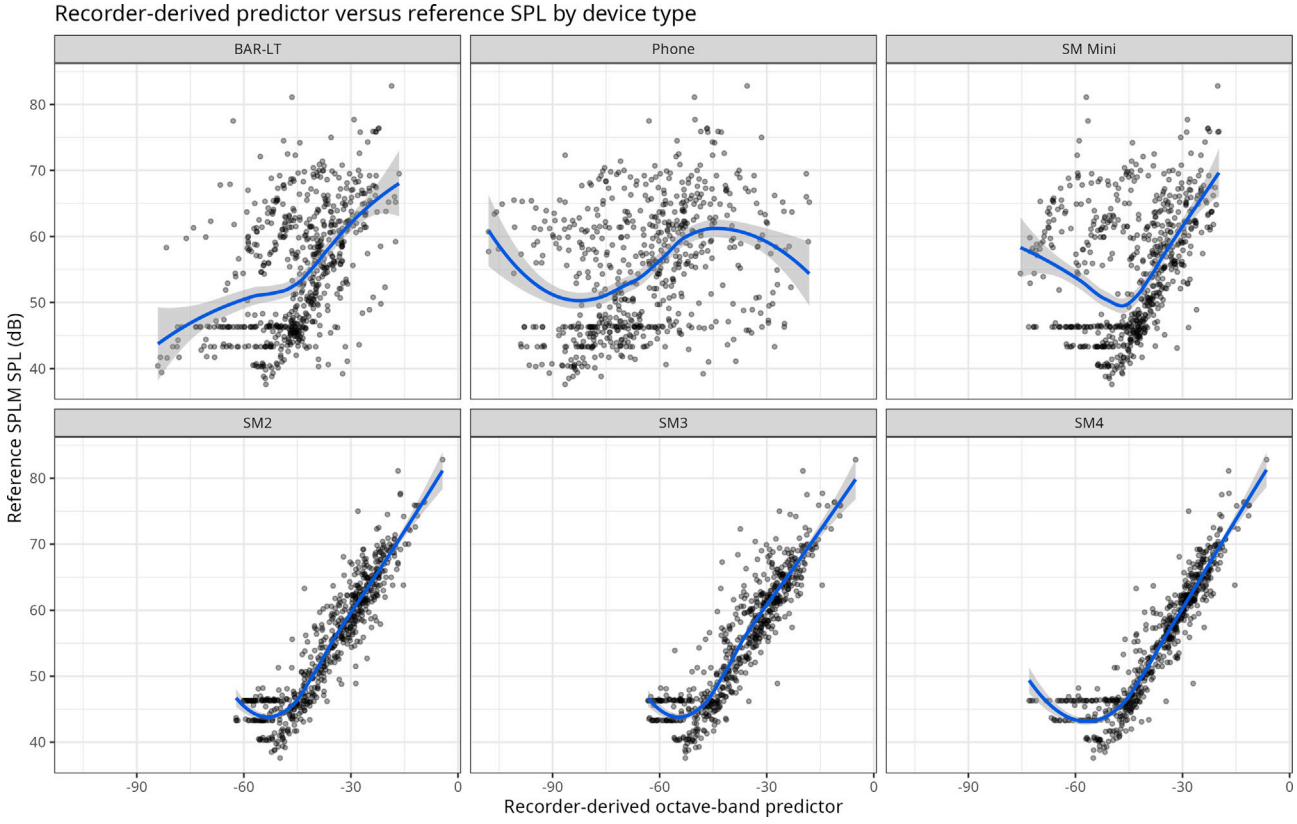


Fig. 3. Recorder-derived octave-band predictor versus reference SPLM SPL by device type. The predictor is the log-transformed octave-band power derived from Welch PSD integration and is not itself an SPL measurement.

Table 1
Out-of-sample predictive performance by device type under 10-fold cross-validation. Bias is the mean residual (SPLM minus estimate), and e denotes the residual. Differences in N among devices reflect a small number of minutes in which one recorder began recording after SPLM logging had already started, so not every minute contributed a complete set of paired device-band observations for every device.

Device	N (band obs.)	Bias (dB)	MAE (dB)	RMSE (dB)	Median $\|e\|$ (dB)	% in ±3 dB	% in ±5 dB	$\%\|e\| \geq 10$ dB
BAR-LT	630	−0.02	3.75	4.75	3.10	49%	71%	3.2%
SM Mini	640	−0.11	3.75	4.66	3.25	46%	71%	2.3%
Phone	640	0.78	4.70	6.08	3.73	42%	63%	11.4%
SM2	640	−0.36	3.44	4.39	2.80	53%	76%	2.3%
SM3	640	−0.36	3.48	4.41	2.90	52%	76%	2.0%
SM4	610	−0.33	3.47	4.37	2.93	51%	76%	2.5%

Table 2
Out-of-sample predictive performance by octave band under 10-fold cross-validation. Bias is the mean residual (SPLM minus estimate).

Band (Hz)	N (device-band obs.)	Bias (dB)	MAE (dB)	RMSE (dB)	% in ±3 dB	% in ±5 dB	% with $\|e\| \geq 10$ dB
16	380	−0.06	4.23	5.60	45%	68%	7.4%
31.5	380	−0.07	3.64	4.53	51%	71%	2.4%
63	380	−0.07	3.59	4.46	49%	73%	2.1%
125	380	−0.12	3.29	4.41	56%	81%	2.9%
250	380	−0.10	4.40	5.33	37%	62%	6.6%
500	380	−0.09	4.72	5.79	33%	62%	7.4%
1000	380	−0.06	4.85	5.77	31%	53%	5.5%
2000	380	−0.03	4.14	5.05	41%	66%	2.9%
4000	380	−0.00	2.74	3.73	66%	87%	2.4%
8000	380	−0.04	2.06	2.52	79%	96%	0.3%

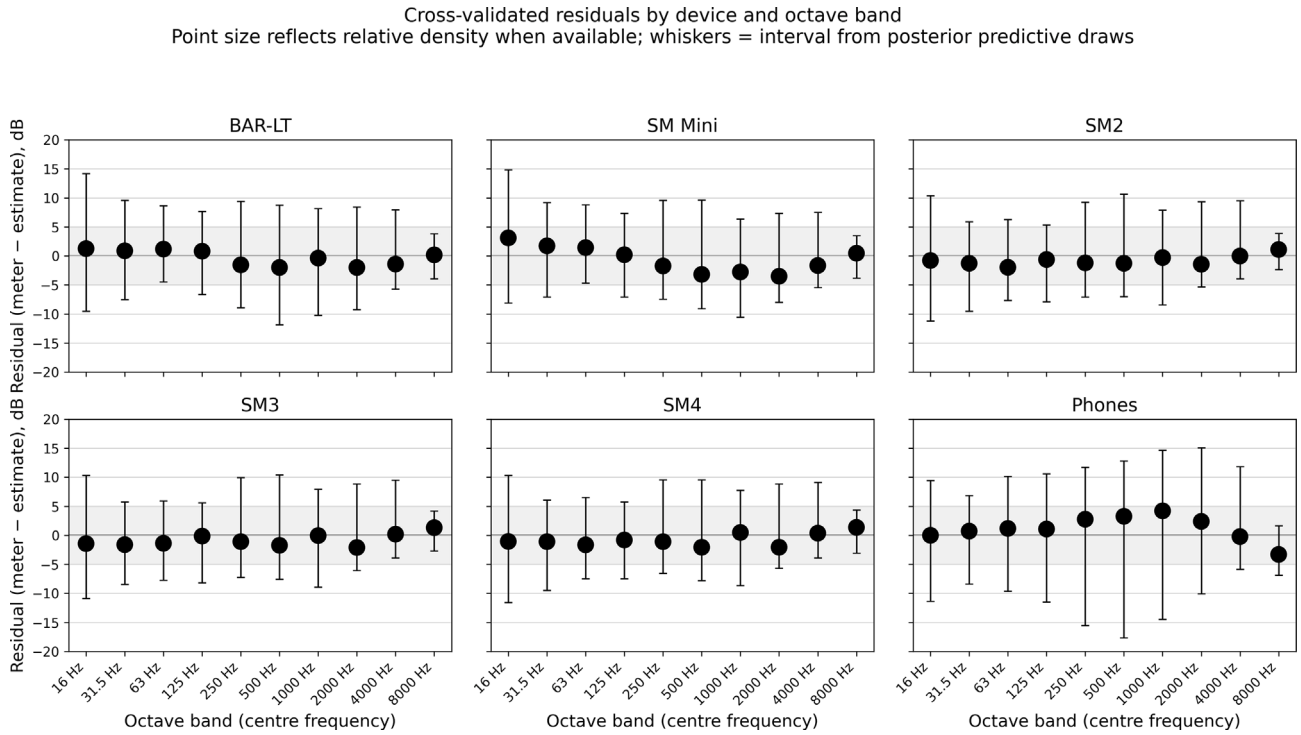


Fig. 4. Cross-validated residuals by device type and octave band. Points summarise residual location by band (SPLM minus estimate), with whiskers indicating intervals from posterior predictive draws; point size reflects relative density when available.

3.5. Performance by octave band

Error varied across octave bands (Table 2), with the smallest RMSE at 8 kHz and higher RMSE in mid-frequency bands. Band-specific performance is expected when device responses and soundscape composition differ across frequencies [4]. RMSE was 2.52 dB at 8 kHz, while 500 Hz and 1 kHz were both close to 5.8 dB RMSE. Band-specific error expectations can be incorporated when aggregating bands or prioritising frequencies relevant to a given application. Residual behaviour by device type and octave band is summarised in Fig. 4, while Fig. 5 provides a visual overview of SPLM SPL versus cross-validated calibrated estimates by device type and octave band.

3.6. Signal-processing sensitivity of the octave-band predictor

We compared the rectangular Welch PSD integration used in the calibration model with a smooth roll-off octave-band approximation for

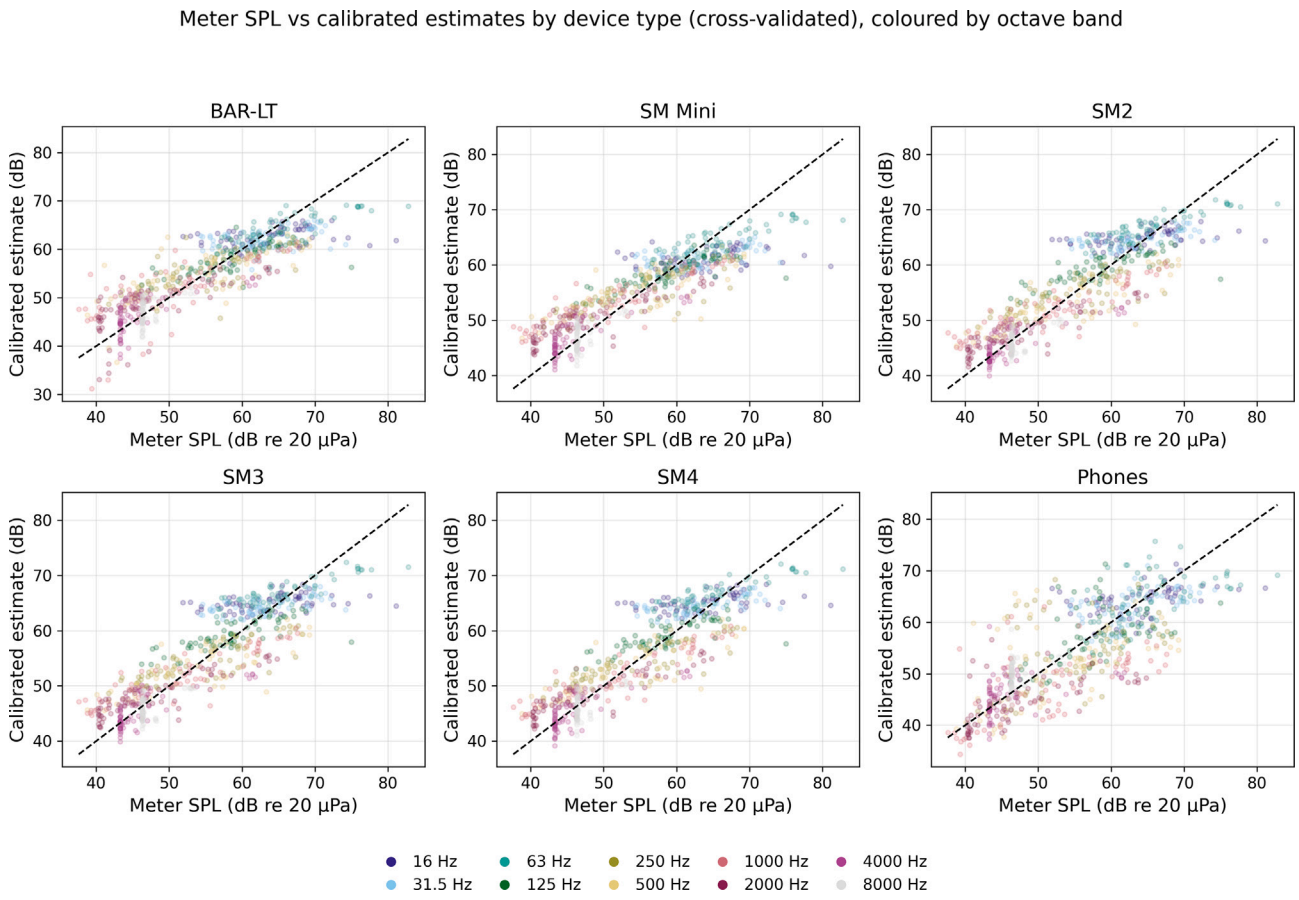


Fig. 5. SPLM SPL versus cross-validated calibrated estimates by device type, coloured by octave band. The dashed line indicates the 1:1 relationship.

the 380 calibration WAV files, yielding 3800 file-band comparisons. This comparison evaluated whether the rectangular integration step materially changes the recorder-derived octave-band predictor. Using a 1/12-octave transition width around the theoretical band edges, the smooth roll-off calculation gave a mean difference of 0.45 dB and a median difference of 0.41 dB relative to rectangular integration (Table 3). Most file-band differences were small, with 96.9% within 1 dB and 99.7% within 2 dB, indicating that the extraction choice produced measurable but generally modest differences relative to the cross-validated calibration error.

3.7. Meteorological domain and sensitivity analyses

The calibration data covered a narrow meteorological domain: temperature ranged from 16.8 to 23.8°C, relative humidity from 37.8 to 52.8%, and wind speed from 0 to 14.8 km/h. These ranges define the empirical domain over which the weather terms were estimated. Applications outside these ranges require extrapolation of the weather coefficients. The coefficient tables should therefore be applied with explicit reference to the observed covariate domain.

Cross-validated performance differed among the primary weather model, the no-weather sensitivity model, and the broad-context sensitivity model (Table 4). This comparison evaluates whether the meteorological terms are required for prediction and whether broad acoustic context explains a similar residual structure. Removing weather modestly reduced performance relative to the primary model, whereas replacing weather with broad acoustic context improved cross-validated performance. These results show that the meteorological covariates are useful operational predictors within the sampled range, but they should not be interpreted as isolated physical weather-response coefficients.

3.8. Calibration coefficients

Posterior summaries for band-specific intercepts α_b and slopes β_b in Eq. (2) are given in Table 5. These coefficients define the core band-specific mapping from device-derived octave-band level to meter-referenced SPL. Applying these coefficients yields a calibrated estimate for each band, conditional on device type and covariates. These coefficients enable direct reuse of the calibration without requiring refitting, provided the same extraction pipeline is used.

Device-type offsets γ_m and covariate coefficients in Eq. (2) are given in Tables 6 and 7. Offsets apply to the correct device identity, and covariate terms provide a plug-in adjustment when sensitivity and weather are recorded and defined consistently. Together with Table 5, these terms define the complete calibration equation for new recordings. This supports routine conversion of recorder-derived octave-band levels to meter-referenced SPL for field monitoring and mapping. Band-specific residual standard deviations are reported in Table 8.

Using the coefficients

To convert a new 60 s device recording into meter-referenced octave-band SPL, follow the steps below. A stepwise workflow reduces ambiguity in coefficient application and keeps new data consistent with the calibration definition of an observation. The steps below yield one calibrated SPL estimate per octave band, with practical error expectations provided by cross-validation tables. This supports immediate operational use without requiring refitting.

1. Use a 60 s recording from a tested device type, or a 60 s window extracted from a longer recording, and retain the calibrated device type m and unit microphone sensitivity s. For analysis we used consecutive non-overlapping 60 s windows to match the SPL meter logs. For application, the window should be processed with the same spectral summarisation workflow, and recordings from untested device types or from units with sensitivity values outside the sampled range should be treated as outside the calibration domain unless additional paired calibration data are collected.
2. For each 60 s window and octave band b, compute the recorder-derived band predictor x_b as follows: estimate a Welch power spectral density for the window (Hann window, 4096-sample segments, 50% overlap), integrate the one-sided PSD over the octave-band edges $(f_c/\sqrt{2},\ f_c\sqrt{2})$ to obtain band power P_b, then set $x_b = 10\log_{10}(P_b)$ and retain finite values for calibration or prediction.
3. Obtain meteorology for the strip, namely T (temperature, °C), H (relative humidity, %), and W (wind speed, km/h), recorded as in the calibration dataset (two reads per strip, with the closest-in-time value assigned to each minute).
4. For each band b, compute

$$\widehat{\mathrm{SPL}}_b = \alpha_b + \beta_b x_b + \gamma_m + \delta s + \theta_T T + \theta_H H + \theta_W W.$$

5. Use Table 2 to attach error expectations by band, and Table 1 to reflect device-type performance.

The intercept α_b is on the SPLM scale (dB re 20 μPa) but is anchored to the recorder-derived predictor scale, where $x_{i,b} = 0$ corresponds to $P_{i,b} = 1$ in PSD-derived units rather than a physically meaningful SPL; α_b therefore completes the linear mapping and is not itself intended to represent a typical field SPL.

Table 3
Signal-processing sensitivity of the recorder-derived octave-band predictor. Differences are smooth roll-off octave-band approximation minus rectangular Welch PSD integration, in dB, for the 380 calibration WAV files. The smooth roll-off comparison used a 1/12-octave transition width around the theoretical octave-band edges, with 4096-sample Welch windows and a 65,536-point diagnostic FFT grid.

Band (Hz)	N	Mean difference	Median difference	95% |difference|	Maximum |difference|
16	380	0.56	0.53	0.79	1.15
31.5	380	0.64	0.54	1.16	1.57
63	380	0.33	0.27	0.90	1.58
125	380	0.52	0.43	1.02	2.06
250	380	0.38	0.35	0.67	1.02
500	380	0.37	0.35	0.60	1.21
1000	380	0.43	0.33	1.03	3.41
2000	380	0.43	0.42	0.76	1.45
4000	380	0.47	0.42	0.87	2.42
8000	380	0.37	0.37	0.62	1.00
Overall	3800	0.45	0.41	0.88	3.41

Table 5
Posterior summaries for band-specific calibration coefficients. For each octave band b, the intercept α_b and slope β_b map the recorder-derived band predictor $x_{i,b}$ (defined in the Methods as $x_{i,b} = 10\log_{10}(P_{i,b})$, where $P_{i,b}$ is octave-band power from PSD integration) to the reference SPLM octave-band SPL for that band (Eq. 2). Values are posterior medians with 95% credible intervals.

Band (Hz)	α_b	α_b 95% CI	β_b	β_b 95% CI
16	100.11	[94.74, 105.58]	0.139	[0.114, 0.165]
31.5	100.03	[94.64, 105.42]	0.166	[0.141, 0.191]
63	104.63	[99.39, 110.05]	0.288	[0.256, 0.319]
125	100.99	[95.60, 106.38]	0.352	[0.312, 0.391]
250	101.85	[96.40, 107.46]	0.443	[0.393, 0.494]
500	103.34	[97.78, 108.92]	0.491	[0.439, 0.543]
1000	100.63	[95.11, 106.11]	0.442	[0.400, 0.484]
2000	94.11	[88.66, 99.53]	0.340	[0.305, 0.374]
4000	90.00	[84.76, 95.26]	0.261	[0.234, 0.287]
8000	85.03	[79.87, 90.24]	0.143	[0.121, 0.165]

4. Discussion

Minute-averaged, field-paired calibration with device-specific terms produced meter-referenced octave-band estimates with an overall RMSE of 4.82 dB across seven contexts. This level of accuracy is informative for acoustic mapping and comparative exposure work where spatial coverage is often more limiting than sub-decibel precision, provided that applications remain within the sampled device, processing, and environmental domains. The protocol provides a practical conversion from recorder-derived spectral summaries to meter-referenced octave-band

Table 4
Cross-validated performance of the primary weather model and two Bayesian sensitivity models. The no-weather model removed temperature, relative humidity, and wind speed. The context model replaced weather covariates with broad acoustic-context offsets. Bias is the mean residual, defined as SPLM SPL minus the calibrated estimate.

Model	N	Bias (dB)	MAE (dB)	RMSE (dB)	Median |e| (dB)	Within ±3 dB (%)	Within ±5 dB (%)	|e| ≥ 10 dB (%)
Primary weather	3800	−0.07	3.77	4.82	3.12	48.6	72.0	4.0
No weather	3800	−0.01	3.95	5.06	3.39	45.2	69.1	4.6
Broad context	3800	−0.03	3.33	4.32	2.65	56.0	78.3	2.8

Table 6
Posterior summaries for device-type offsets γ_m in Eq. (2). Offsets are additive and interpreted relative to BAR-LT, whose offset is fixed at zero for identifiability. Values are posterior medians with 95% credible intervals.

Device	Offset (dB)	95% CI
BAR-LT	0.00	[0.00, 0.00]
SM Mini	3.66	[1.57, 5.73]
Phone	7.53	[6.17, 8.90]
SM2	−1.31	[−1.78, −0.84]
SM3	2.41	[0.66, 4.15]
SM4	−0.30	[−0.91, 0.31]

Table 7
Posterior summaries for covariate coefficients shared across octave bands in Eq. (2). Values are posterior medians with 95% credible intervals. Predictors are retained on their recorded scales to support plug-in use.

Term	Estimate	95% CI
Microphone sensitivity coefficient (δ)	−0.121	[−0.181, −0.061]
Relative humidity coefficient (θ_H)	−0.288	[−0.337, −0.240]
Temperature coefficient (θ_T)	−1.047	[−1.183, −0.916]
Wind speed coefficient (θ_W)	0.103	[0.068, 0.139]

Table 8
Posterior summaries for band-specific residual standard deviations σ_b (dB) in Eq. (2). Values are posterior medians with 95% credible intervals.

Band (Hz)	σ_b	95% CI
16	5.35	[4.98, 5.76]
31.5	4.29	[4.01, 4.62]
63	4.19	[3.91, 4.52]
125	4.16	[3.87, 4.48]
250	5.07	[4.72, 5.46]
500	5.59	[5.21, 6.02]
1000	5.53	[5.15, 5.96]
2000	4.80	[4.47, 5.19]
4000	3.47	[3.24, 3.73]
8000	2.39	[2.17, 2.64]

SPL, with error distributions that can be carried into downstream analyses. This supports ARU networks in reporting exposure estimates on a common SPL scale while making uncertainty explicit, without treating ARUs or smartphones as substitutes for certified sound level meters. Residual diagnostics showed that prediction errors changed across the SPL range, with systematic underestimation for the small number of observations above 70 dB. This matters because high-SPL events may involve different source types, stronger low-frequency components, or recorder acquisition behaviour that is weakly represented by the current field-paired dataset. The overall cross-validation metrics therefore summarise performance across the complete sampled dataset, but they do not imply equal accuracy at the highest observed SPLM values. Applications focused on high-level industrial, transport, or impulsive sound should include additional paired calibration data in that SPL range, and future models could evaluate piecewise or nonlinear terms once such data are available.

The present diagnostics identify the high-SPL pattern but do not isolate its physical cause. Underestimation at high reference levels could arise from sparse training data in that range, low-frequency source structure, context-specific events, dynamic-range behaviour in recorder acquisition chains, or a combination of these mechanisms. We therefore treat high-SPL underestimation as an empirical limitation of the current calibration rather than attributing it to a single device mechanism. A targeted calibration design with controlled source levels and more observations above 70 dB would be needed to distinguish sparse-data effects from nonlinear recorder response. Predictive performance differed by device type, with smartphones showing larger bias and heavier-tailed errors than purpose-built field recorders in this dataset. The predictor-response pattern for smartphones in Fig. 3 was also weaker and appeared more level-dependent than for the better-performing purpose-built recorders. The smartphone recordings were made through the ordinary device pathway, without disabling gain control, compression, or other device-level processing. The pattern in Fig. 3 is therefore compatible with level-dependent gain or dynamic processing such as automatic gain control or compression, although these processes were not experimentally manipulated or measured and the present data cannot identify the mechanism responsible for the observed relationship. The smartphone results should therefore be reported separately from purpose-built ARUs and treated as exploratory for this practical recording pathway. In particular, ordinary smartphone recordings should not be assumed to follow a stable linear transfer relationship across sound levels without device- and acquisition-pathway-specific validation. The linear calibration equation is an empirical predictor for the observed smartphone recordings, not a physical model of the phone gain path or evidence that ordinary smartphones provide stable calibrated acoustic measurements. The smartphone units also showed the widest measured sensitivity range among tested device types (Table C.2), reinforcing that these results should be interpreted as direct-use empirical performance for the tested units rather than as stable electroacoustic sensitivity estimates.

Device-type offsets were estimated relative to BAR-LT and indicated additive differences among several tested device types. These offsets account for persistent device-type differences after conditioning on the band-specific recorder predictor, unit microphone sensitivity, and meteorological covariates. Device type and unit microphone sensitivity therefore provide complementary information for calibration, with device type capturing systematic differences among acquisition pathways and sensitivity capturing measured unit-level variation. This supports recording both the nominal device type and the measured microphone sensitivity when quantitative SPL estimation is an objective.

Error varied across frequencies, with stronger performance at higher frequencies and larger errors in mid-frequency bands, including 500 Hz and 1 kHz. Frequency-dependent behaviour is expected when recorder responses and soundscape composition differ across bands [4]. Retaining octave-band structure supports targeted interpretation and provides a basis for prioritising refinement in bands that dominate uncertainty for the intended application. Future expansions of the dataset can focus on contexts and bands where errors remain greatest, while preserving a stable application workflow.

The calibration model estimated band-specific intercepts, slopes, and residual standard deviations while sharing device-type offsets and covariate effects across bands. It did not explicitly model residual covariance among neighbouring bands or use the full spectral shape of each soundscape as a joint predictor. This is a simplification because many environmental sound sources are characterised by spectral profiles rather than by isolated octave-band levels. The present equations should therefore be interpreted as band-specific empirical translations from recorder-derived predictors to meter-referenced SPLM values, not as a multivariate source-separation or spectral-matching model. Future work could extend this framework using multivariate band models, hierarchical smoothing across frequency, or source-aware spectral-shape models when the objective requires inference about sound-source identity or covariance among bands.

The univariate linear model was chosen because it yields directly applicable per-band coefficients and separates the operational calibration problem from more complex source-classification or spectral-reconstruction tasks. This choice is appropriate for a modest field-paired dataset whose main objective is to estimate meter-referenced octave-band SPL with quantified error rather than to identify sound sources or infer complete spectral covariance. Alternative models, including nonlinear calibration curves, piecewise high-SPL models, multivariate spectral models, and context-aware hierarchical models, may improve

prediction when larger calibration datasets are available. The sensitivity analyses and residual diagnostics added here identify where those extensions are most likely to matter, particularly for high-SPL observations, acoustic-context structure, and spectral dependence among bands.

Two design choices in this protocol are the use of 60 s averages and the restriction to octave bands centred at frequencies up to 8 kHz. These choices keep the recorder-derived summaries aligned with the SPLM logging interval and avoid inconsistent definitions near the upper limit of common recorder sampling rates, which would otherwise affect the 16 kHz band. Minute averaging stabilises paired summaries, and the octave-band restriction keeps the calibration tied to frequency bands that are consistently represented across the tested devices and processing workflow. These choices prioritise portability and consistency for defined field applications, while leaving finer temporal or spectral resolution to future analyses when explicitly required.

Coefficients were estimated on recorded predictor scales rather than standardised scales [4]. This supports plug-in use with recorded values of sensitivity and meteorology, but it means coefficient magnitudes are not intended for direct comparison across predictors. Interpretation should therefore focus on predictive performance and on the protocol's operational application rather than on ranking covariate influence. This keeps the calibration aligned with its role as a practical translation tool within a monitoring workflow.

The weather sensitivity analysis showed that removing meteorological covariates modestly reduced predictive performance, whereas replacing them with broad acoustic-context offsets improved prediction. This indicates that the weather terms contain useful predictive information within the sampled data, but they may also partly represent context, site, source composition, or deployment conditions. We therefore interpret the temperature, humidity, and wind coefficients as empirical calibration terms within the observed covariate domain, not as direct physical estimates of microphone or propagation response. Because context labels are study-specific and may not transfer cleanly to new deployments, the context model is most useful as evidence of residual acoustic-context structure rather than as the primary application equation. Future calibration datasets should deliberately expand seasonal and meteorological coverage so that weather effects can be separated more clearly from acoustic context and deployment structure.

The signal-processing sensitivity analysis showed that replacing rectangular Welch PSD integration with a smooth roll-off octave-band approximation changed most recorder-derived band predictors by less than 1 dB. This indicates that the rectangular integration choice contributes measurable processing uncertainty, but that this uncertainty is generally smaller than the cross-validated prediction error of the calibration model. We therefore retain the original extraction pipeline for coefficient application, while making explicit that the coefficients are conditional on that pipeline. Applications requiring formal standards-compliant fractional-octave measurements should use a validated filtering implementation and should calibrate or validate that workflow separately against the reference meter.

This paper provides both coefficient tables and the calibration workflow that defines how SPLM and recorder summaries are paired, summarised, and modelled. Because calibration is empirical, the current coefficients are bounded by the sampled contexts, weather range, device types, SPL range, and audio-extraction pipeline. The present dataset should therefore be treated as an initial field-paired calibration for defined operating conditions, not as a universal calibration across all seasons, source types, SPL regimes, recorder settings, or consumer devices. In particular, the smartphone results should not be generalised beyond the tested units and acquisition pathway because uncontrolled level-dependent processing may violate the approximately linear relationship assumed by the calibration model. The 64 minute-level windows should be viewed as field snapshots of the sampled sound contexts rather than as a complete representation of the temporal variance of airport, construction, transport, industrial, park, backyard, or road soundscapes. Applying the same workflow to expanded datasets will refine the coefficient estimates, test transferability, and support a cumulative approach in which ARU deployments produce increasingly consistent octave-band SPL estimates with well-characterised uncertainty.

CRediT authorship contribution statement

Alberto De Rosa: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Apoorv Saini:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Hedwig Lankau:** Writing – review & editing, Resources, Project administration. **Matina Kalcounis-Rueppell:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Erin Bayne:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Data and code availability

The analysis-ready tabular data supporting this article are available from Zenodo at https://doi.org/10.5281/zenodo.19959797. These files provide the recorder-derived octave-band summaries, sound pressure level meter reference measurements, microphone sensitivity measurements, meteorological covariates, README file, and a data dictionary needed to reproduce the calibration models and cross-validation summaries reported here. The data processing, modelling, signal-processing sensitivity scripts, and manuscript source files are available from the project GitHub repository at https://github.com/baynelab-research/ARUtoSPL. Raw audio files are not deposited as part of the public data package, so the deposited tabular data support direct reproduction of the calibration analyses, while the repository documents the waveform-processing steps needed to apply or validate the extraction workflow in new datasets.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI ChatGPT to improve readability and to help improve, clean, and comment on analysis code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Combined coefficients by band and device type

For convenience, Table 9 lists the combined intercept ($\alpha_b + \gamma_m$) and slope β_b for each band and device type, using posterior medians. Some users prefer a single per-band, per-device intercept and slope for implementation in simple scripts, while still applying covariate adjustments separately. Users should still apply the shared covariate terms in Eq. (2) when sensitivity and meteorology are available. Providing both representations supports reuse across varied implementation preferences.

Table 9
Combined calibration coefficients for immediate application. For each octave band and device type, the intercept is $\alpha_b + \gamma_m$ and the slope is β_b from Eq. (2), using posterior medians. Covariate terms ($\delta s + \theta_T T + \theta_H H + \theta_W W$) should be added separately.

Band (Hz)	Device	Intercept (dB)	Slope
16	BAR-LT	100.11	0.139
16	SM Mini	103.78	0.139
16	Phone	107.62	0.139
16	SM2	98.80	0.139
16	SM3	102.53	0.139
16	SM4	99.81	0.139
31.5	BAR-LT	100.03	0.166
31.5	SM Mini	103.67	0.166
31.5	Phone	107.56	0.166
31.5	SM2	98.72	0.166
31.5	SM3	102.41	0.166
31.5	SM4	99.73	0.166
63	BAR-LT	104.63	0.288
63	SM Mini	108.27	0.288
63	Phone	112.13	0.288
63	SM2	103.32	0.288
63	SM3	107.03	0.288
63	SM4	104.33	0.288
125	BAR-LT	100.99	0.352
125	SM Mini	104.63	0.352
125	Phone	108.50	0.352
125	SM2	99.69	0.352
125	SM3	103.39	0.352
125	SM4	100.69	0.352
250	BAR-LT	101.85	0.443
250	SM Mini	105.53	0.443
250	Phone	109.39	0.443
250	SM2	100.56	0.443
250	SM3	104.30	0.443
250	SM4	101.58	0.443
500	BAR-LT	103.34	0.491
500	SM Mini	106.97	0.491
500	Phone	110.84	0.491
500	SM2	102.03	0.491
500	SM3	105.73	0.491
500	SM4	103.04	0.491
1000	BAR-LT	100.63	0.442
1000	SM Mini	104.27	0.442
1000	Phone	108.14	0.442
1000	SM2	99.32	0.442
1000	SM3	103.03	0.442
1000	SM4	100.32	0.442
2000	BAR-LT	94.11	0.340
2000	SM Mini	97.70	0.340
2000	Phone	101.58	0.340
2000	SM2	92.81	0.340
2000	SM3	96.48	0.340
2000	SM4	93.81	0.340
4000	BAR-LT	90.00	0.261
4000	SM Mini	93.62	0.261
4000	Phone	97.50	0.261
4000	SM2	88.70	0.261
4000	SM3	92.36	0.261
4000	SM4	89.69	0.261
8000	BAR-LT	85.03	0.143
8000	SM Mini	88.64	0.143
8000	Phone	92.52	0.143
8000	SM2	83.72	0.143
8000	SM3	87.41	0.143
8000	SM4	84.74	0.143

Appendix B. Worked application examples

This appendix illustrates the complete workflow from waveform to calibrated SPL estimate using two observed calibration rows whose covariates and recorder-derived predictors fall inside the empirical training domain. Using in-domain examples avoids presenting coefficient application as being valid outside the sampled meteorological, device-sensitivity, or recorder-predictor ranges. The examples show how the coefficients in Tables 5–7 are combined for one band and one device at a time. They provide a reproducible application recipe while preserving the condition that the calibration is defined by the sampled data and the stated extraction pipeline.

To illustrate the complete workflow from waveform to calibrated SPL estimate, we provide two worked examples using observed 60 s recordings and covariate values from the calibration dataset. In both cases, the WAV recording was processed using the same spectral pipeline described in the Methods (Welch PSD with Hann window, 4096-sample segments, 50% overlap), implemented in R using the `tuneR` and `seewave` packages [8,9].

Example 1: SM4 unit, 1 kHz band. The first example uses the observed file `backyard_2_sm4-01.wav`, corresponding to an SM4 recording from the backyard context, strip 2, recording 1. This row was selected because its meteorological values, microphone sensitivity, and recorder-derived 1 kHz predictor are all inside the observed training ranges. It provides an in-domain example of the 1 kHz calibration equation for an SM4 unit. The calculation demonstrates coefficient application without requiring extrapolation beyond the calibration dataset.

The Welch power spectral density was estimated with 4096-sample Hann windows and 50% overlap. The one-sided PSD was integrated numerically between the 1 kHz octave-band edges

$$\left(\frac{1000}{\sqrt{2}},\ 1000\sqrt{2}\right) = (707.1,\ 1414.2)\ \text{Hz}$$

to obtain band power

$$P_{1000} = 3.56 \times 10^{-5}.$$

The recorder-derived predictor is then

$$x_{1000} = 10\log_{10}(P_{1000}) = 10\log_{10}(3.56 \times 10^{-5}) = -44.48.$$

For this observed SM4 row, microphone sensitivity was $s = -32.7$ dBV/Pa. Meteorological values aligned with this minute were $T = 20.9°\text{C}$, $H = 43.8\%$, and $W = 0$ km/h.

From Table 5 (1 kHz band):

$$\alpha_{1000} = 100.63, \quad \beta_{1000} = 0.442.$$

From Table 6 (SM4):

$$\gamma_{\text{SM4}} = -0.30.$$

From Table 7:

$$\delta = -0.121, \quad \theta_T = -1.047, \quad \theta_H = -0.288, \quad \theta_W = 0.103.$$

Substituting into Eq. (2):

$$\widehat{\text{SPL}}_{1000} = 100.63 + 0.442(-44.48) - 0.30 + (-0.121)(-32.7) + (-1.047)(20.9) + (-0.288)(43.8) + 0.103(0).$$

Evaluating each term using the rounded coefficients reported in the tables:

$$100.63 - 19.66 - 0.30 + 3.96 - 21.88 - 12.61 + 0.00 = 50.13\ \text{dB}.$$

Thus, the calibrated 1 kHz SPL estimate for this 60 s window is 50.1 dB.

Example 2: BAR-LT unit, 4 kHz band. The second example uses the observed file `backyard_2_barlt-01.wav`, corresponding to a BAR-LT recording from the same context, strip, and recording minute as Example 1. Using the same minute with a different device and band keeps the meteorological values fixed while showing how the device and band coefficients change the calculation. It provides an in-domain example of the 4 kHz calibration equation for a BAR-LT unit. The example shows how the same workflow is applied across device types and octave bands.

The 4 kHz octave band has edges

$$\left(\frac{4000}{\sqrt{2}}, 4000\sqrt{2}\right) = (2828.4,\ 5656.9)\ \text{Hz}.$$

Numerical integration of the Welch PSD over this interval yielded

$$P_{4000} = 2.69 \times 10^{-5}.$$

Hence,

$$x_{4000} = 10\log_{10}(2.69 \times 10^{-5}) = -45.70.$$

For this observed BAR-LT row, microphone sensitivity was $s = -41.425$ dBV/Pa, and concurrent meteorology was $T = 20.9$°C, $H = 43.8\%$, and $W = 0$ km/h.

From Table 5 (4 kHz):

$$\alpha_{4000} = 90.00, \quad \beta_{4000} = 0.261.$$

From Table 6 (BAR-LT):

$$\gamma_{\text{BAR-LT}} = 0.00.$$

From Table 7:

$$\delta = -0.121, \quad \theta_T = -1.047, \quad \theta_H = -0.288, \quad \theta_W = 0.103.$$

Substituting into Eq. (2):

$$\widehat{\text{SPL}}_{4000} = 90.00 + 0.261(-45.70) + 0.00 + (-0.121)(-41.425) + (-1.047)(20.9) + (-0.288)(43.8) + 0.103(0).$$

Evaluating each term using the rounded coefficients reported in the tables:

$$90.00 - 11.93 + 0.00 + 5.01 - 21.88 - 12.61 + 0.00 = 48.59\ \text{dB}.$$

The calibrated 4 kHz SPL estimate for this minute is therefore 48.6 dB.

In both examples, the predictor x_b was computed using the identical spectral integration procedure described in the Methods, and the covariate values fall within the observed calibration ranges reported in Tables C.1 and C.2. Performance expectations for each device type and band are given in Tables 1 and 2.

Appendix C. Covariate summaries

Table C.1
Overall meteorological covariates recorded during paired calibration sampling.

Covariate	Mean	Min	Max
Temperature (°C)	21.16	16.8	23.8
Relative humidity (%)	45.49	37.8	52.8
Wind speed (km/h)	3.39	0.0	14.8

Table C.2
Microphone sensitivity summaries by device model.

Device model	N units	Sensitivity (mean [min, max])
BAR-LT	3	−40.05 [−41.42, −39.25]
SM Mini	3	−6.14 [−6.45, −5.96]
SM2	3	−39.03 [−40.55, −36.85]
SM3	3	−11.42 [−14.65, −8.35]
SM4	3	−33.38 [−33.85, −32.70]
Phone	3	−19.80 [−26.16, −15.84]

Data availability

Research link provided.

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